

Townsend's 2026 Monthly Dust Loss Model

Purpose

This model is used to estimate the percentage of potential monthly photovoltaic (PV) energy sacrificed due to dust soiling.

PV financing decisions are based on long-term energy estimates, and popular PV modeling software such as PVsyst include monthly inputs for soiling losses – snow in the winter, dust in the summer. This modeling tool relies principally on monthly rainfall to determine a changing pattern of soiling throughout the year. That pattern is also influenced by four other factors:

1. The daily soiling ramp rate (seasonally adjustable).
2. The number of manual array washings, up to two per year.
3. The influence of snow, where any month with at least a 5% projected snow loss will govern. This level of snow loss will treat that month's dust soiling loss as zero.¹
4. For bifacial arrays, the back side's contribution is presumed to have no appreciable dust buildup. The full array's monthly dust loss is reduced a little by the percentage of energy contributed from the always-clean back side.

Rules

The following table lists the core assumptions applied to each month, subject to the superseding influence of snow, which sets the dust loss to 0% if that month's snow losses are at least 5%. For months with snow less than 5%, the total loss = snow + dust – (snow x dust).

Monthly precipitation, inches (mm)	Soiling
≥4 (102)	0%
≥2 and <4 (51-102)	1%
≥0.5 and <2 (13-51)	If prior month precip. ≥2", add 8 days x seasonal ramp rate to prior month's soiling. If prior month precip. <2", add 15 days x seasonal ramp rate to prior month's soiling.
<0.5 (13)	Add product of number of days per month x seasonal ramp rate applicable to current month to prior month's soiling.

¹ Townsend's snow loss measurements during 2013-15 for seven orientations at Golden, CO yielded six months with one snow event. Over these 42 qualifying data points, the average monthly snow loss was 4.8%, so for this tool, we have rounded the mixed dust and snow threshold to 5%. This is viewed as the typical snow loss for which one snow event per month has occurred, and that one snow event is viewed as sufficient to clean the dust off. The average inches of snow for the one-event months was 4 inches; again, this is viewed as a sufficient amount of snow to clean off any dust.

A few other rules are used, too. 1) A maximum monthly loss of 30% is assumed. 2) A starting month must be chosen. We need a consistent, wrapped pattern of monthly dust losses applicable to all years. To do this, the month with the most precipitation is chosen as the starting month. The soiling applicable to that month is subject to the same rules described above, with one exception: if the starting month's precipitation is less than 2 inches, the soiling for that month is set to 2%. 3) A manual wash is assumed to occur on the first day of a month. Unlike the Kimber et al. modelⁱ, this model does not include a "grace period" of 0% soiling after a manual wash. The dust loss in a manual wash month will still be calculated based on a half month of accumulation at the daily ramp rate applicable to that season.

If one or two manual washings per year are selected, the model does an exhaustive search, testing each month in sequence as the date of the first potential wash, then recomputing the *annual* energy loss associated with each potential choice. After checking each month, the wash month with the lowest annual soiling loss is deemed to be optimal. The same logic is used to determine when best to do a second wash. It is possible that the second-best time to do a manual wash may occur chronologically in a month earlier than the best first wash. **The incremental annual loss reduction value of the "second" wash is predicated on committing to also performing the more impactful "first" wash in its prescribed month.** Here, the labeling of "best" and "second-best" is more important than the numbering of the months. If wash #1 is prescribed for August and wash #2 is prescribed for July, the low annual soiling loss being sought for two manual washes will only be attained if both the July and then the August washings occur.

Inputs

1. Twelve monthly precipitation amounts, specified in units of inches or millimeters.
2. Four seasonal daily dust accumulation (ramp) rates, with four rate choices per season, with summer and fall periods tending to show higher ramp rates than in the spring or winter:
 - a. 0.025%, for ultra-sandy, low moisture, inorganic particulates, <10"/yr rain
 - b. 0.05%, scrub desert, rain between 10-15"/yr
 - c. 0.1%, typical default, safely applicable in many climates
 - d. 0.15%, humid, agriculturally active, sooty, pollens, or prone to bird droppings
3. Number of manual washes, from 0 to 2 per year.
4. Bifacial: If bifacial, a separate calculation linked to the snow model is used to determine the monthly percentage of generation attributable to the essentially unsoiled back side, with the overall dust loss reduced accordingly.

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ⁱ Kimber, A., Mitchell, L., Nogradi, S. and Wenger, H. (2006) The Effect of Soiling on Large Grid-Connected Photovoltaic Systems in California and the Southwest Region of the United States. Proceedings of the 4th IEEE World Conference on Photovoltaic Energy Conference, Waikoloa, 2391-2395.
<https://doi.org/10.1109/WCPEC.2006.279690>